

Supervised Fine-Tuning (SFT) and Alignment in Large Language Models

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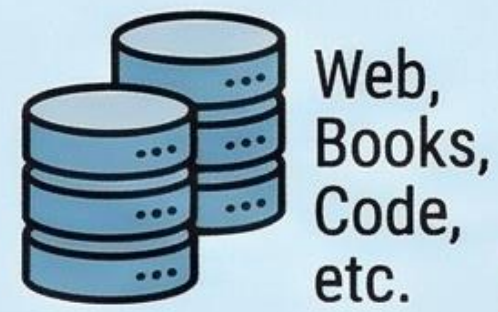
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Agenda

1. The Foundation: Transformer Architecture & Training Paradigms
2. From Raw Models to Useful Assistants: Instruction Tuning
3. Human Alignment: RLHF, PPO, and DPO
4. Mixture of Experts Models
5. Reasoning Models: Chain-of-Thought and Beyond

1. DATA PREPARATION



Raw Data Sources

Web, Books, Code, etc.

Data Cleaning

- Filtering
- De-duplication
- Formatting
- Tokenization

Cleaned Pretraining Corpus

2. PRETRAINING



Pretraining

- Self-Supervised Learning on Corpus
- Learns Language Patterns, Facts, & Reasoning

Base Model
(e.g., Llama-3-Base, GPT-3-Base)

3. POST-TRAINING (ALIGNMENT & FINE-TUNING)



Instruction Tuning

- Supervised Fine-Tuning (SFT) on Instruction-Response Pairs
- Learns to Follow Instructions

Instruction-Tuned Model



Policy Optimization Reinforcement Learning

- Collect Preference Data
- Train Reward Model
- Optimize Model with PPO/DPO
- Max Human Preference Scores

Aligned Model

e.g., Llama-3-Instruct, ChatGPT

4. INFERENCE & APPLICATION



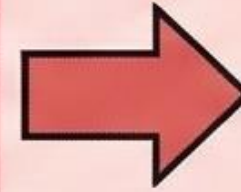
User Prompt

Prompting & Interaction Techniques

In-Context Learning (ICL)
Examples, few-shot

Chain of Thought (CoT)
Step-by-step

Tree of Thought (ToT)
Explore possibilities



Inference

Generating

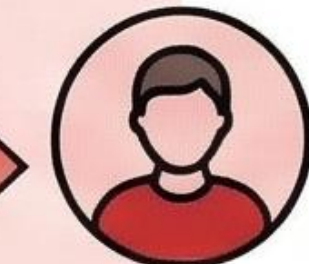
RAG

(Retrieval-Augmented Generation)

Retrieve Relevant Docs



External Knowledge Base



Final Response

5.

Security Evaluation & Red Teaming



Security Evaluation & Red Teaming

- Test against Inversion Attacks (Model/Prompt Inversion)
- Analyze to Recover Prompts or Training Data
- Post-Deployment Evaluation

Foundation vs Instruction Tuned Models

Foundation

- . Coherent, fluent generation
- . Completion and continuation
- . Vast world knowledge
- . Potentially harmful outputs
- . Difficult to control
- . Misaligned with our values

Instruction Tuned

- . Foundation ability and more
- . Controllable by prompt
- . Less need to fine-tune
- . Follows any command 
- . Slight hit to performance
- . Sensitive to prompt template

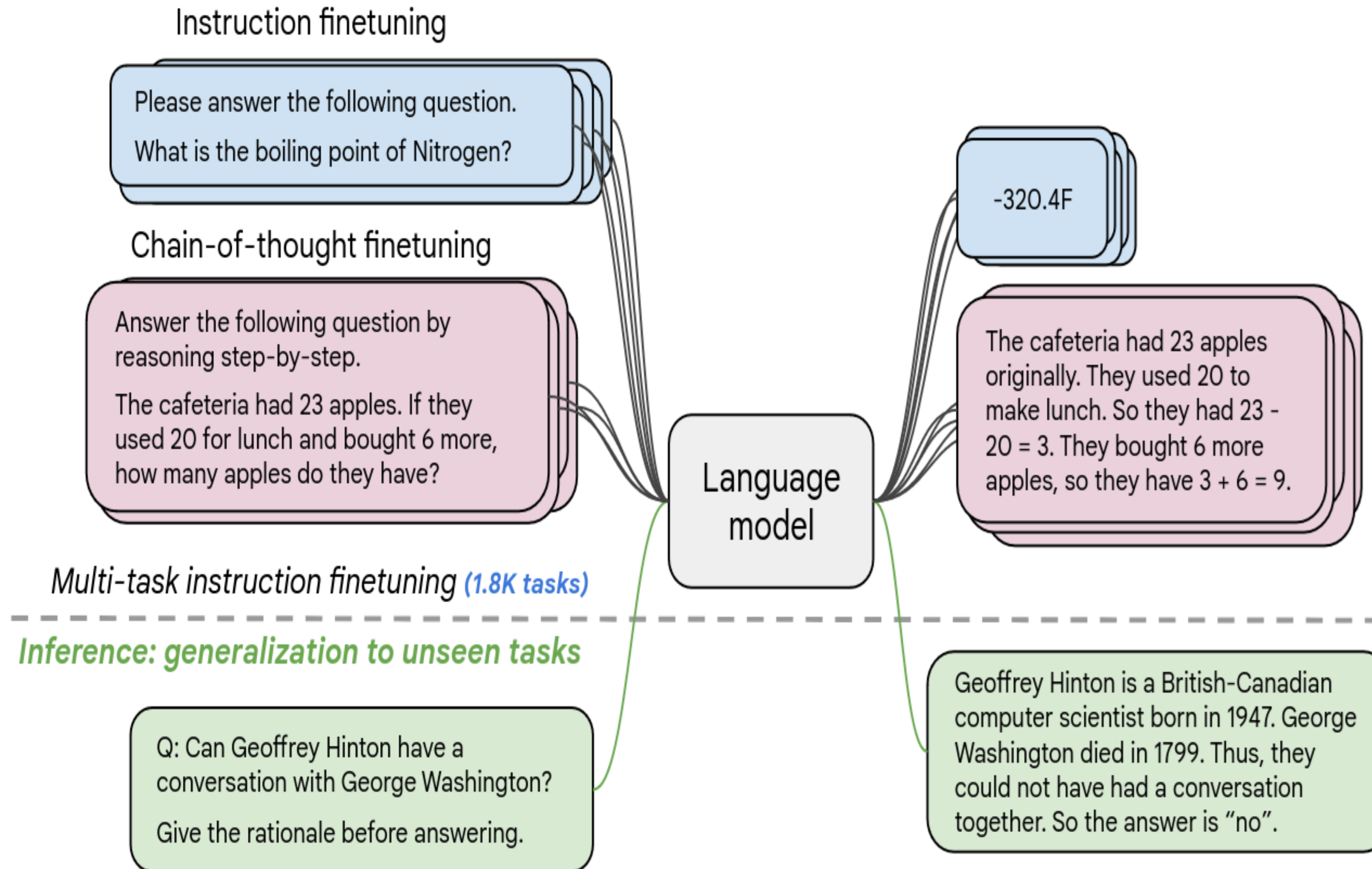
Instruction Tuning

- Before Instruction Tuning:
 - Trained to minimise next-token prediction loss on large corpora
 - Not explicitly trained to follow user instructions
- Motivation for Instruction Tuning:
 - Reduce mismatch between training objective and user intent
 - Used to provide useful information and reliably respond to instructions

Discussion Question

- If you deployed a raw foundation model directly in customer support, what kinds of failures would you expect, even if the language sounds fluent?

Instruction Tuning



HHH - Guiding Principles

- **Helpful**

- Comprehend user's intentions
- Execute requested actions correctly
- Offers relevant supporting information and alternate solutions

- **Honest**

- Provide truthful, meaningful and specific information
- Admits when it is unable to produce relevant or correct information
- Advertises when a hypothesis is presented instead of fact

- **Harmless**

- Generated content does not contain offensive terminology or ideology
- Avoids providing information on potentially dangerous activities
- Refuses to participate in unlawful activities
- Can detect attempts to deceive and manipulate to extract illicit information

Prompt Template

```
<s>[INST] <<SYS>>
{{ system_prompt }}
<</SYS>>

{{ user_message }} [/INST]
```

[6] LLaMa2: Open Foundation and Fine-Tuned Chat Models

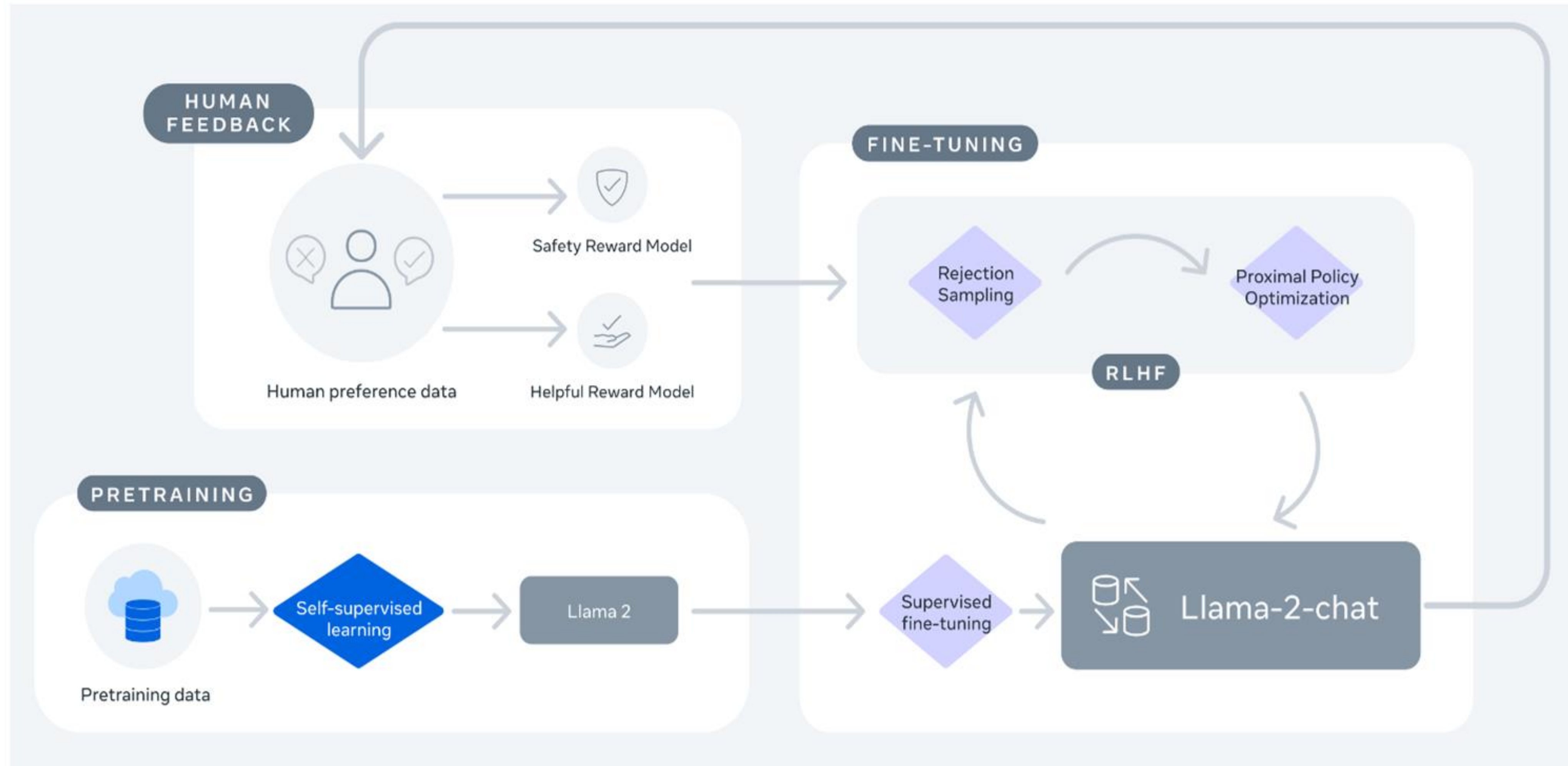
```
<|im_start|>system
{system_message}<|im_end|>
<|im_start|>user
{prompt}<|im_end|>
<|im_start|>assistant
```

[7] Qwen Technical Report

Instruction: {{instruction}}
Output:

[8] Phi-2: The surprising power of small language models

Reinforcement Learning with Human Feedback (RLHF)



Reward Modelling

- Binary pairwise ranking loss [9]
 - $r_{\theta}(x, y)$ is the scalar reward output for prompt x and response y
 - y_c is the chosen response in the pairwise matchup, y_r is the rejected
- Cross entropy: the difference in reward represents the log odds that one response is preferred by human labellers.

$$\mathcal{L}_{\text{ranking}} = -\log(\sigma(r_{\theta}(x, y_c) - r_{\theta}(x, y_r)))$$

Discussion question

- Why might pairwise comparison be easier and more reliable than asking humans to give every answer an absolute score from 1 to 10?

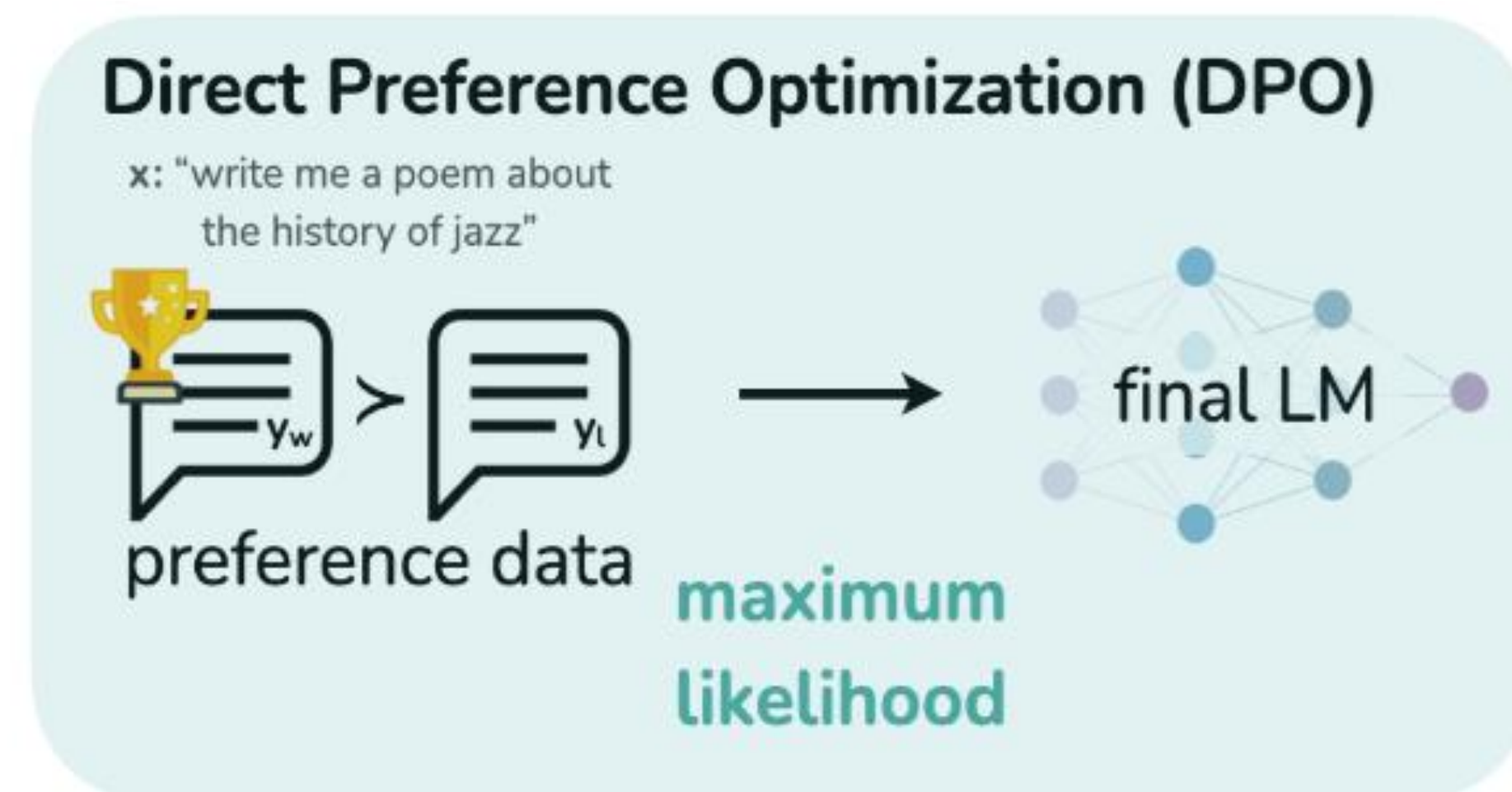
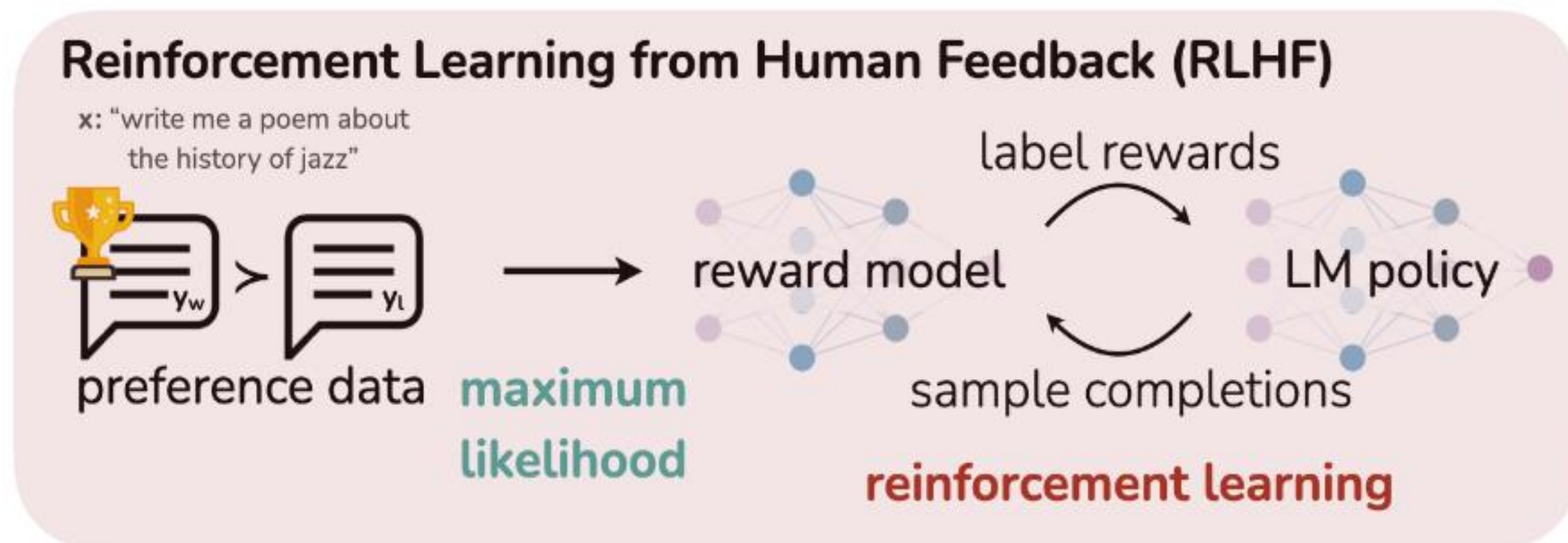
Proximal Policy Optimisation (PPO)

- Motivation: How can we take the biggest improvement step on a policy using the currently available data, without stepping so far we cause collapse?
 - Small policy steps often result in convergence but takes a long time
 - Large steps may converge faster but may collapse irrecoverably

Ratio Function	$r_t(\theta) = \pi_{\theta}(a_t s_t) / \pi_{\theta_{old}}(a_t s_t)$
Advantage	$A_t = \text{reward} - \text{baseline}$
Clipping	Prevents destructive updates

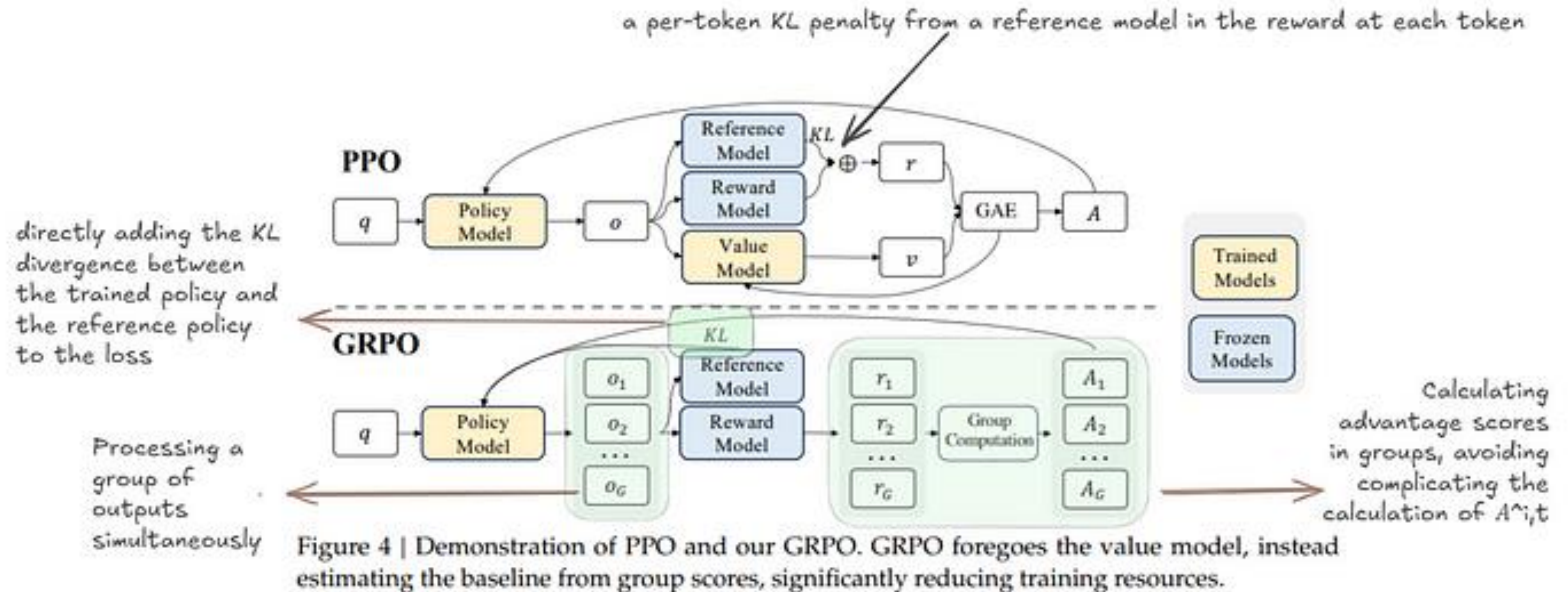
Direct Preference Optimisation

- Rafailov et al. argue that the reward modelling approach is too complex and unstable [13]
- They propose to replace the RLHF procedure with a direct, exact optimisation using binary cross entropy



[13] Direct Preference Optimization: Your Language Model is Secretly a Reward Model

Group Relative Policy Optimisation



- Reward model is replaced with a series of rule based rewards as python functions which proxy desired behaviour
- Averaged reward from multiple generations eliminates need for value model
- Processes multiple outputs for a given prompt simultaneously
- Advantage scores are calculated relative to the group
 - Aligns with original reward model objective

GRPO vs PPO

GRPO

- . No value model needed
- . Higher sample efficiency
- . Functional rewards
- . Less studied and optimised
- . Sensitive to batching
- . Quite expensive (2 models)

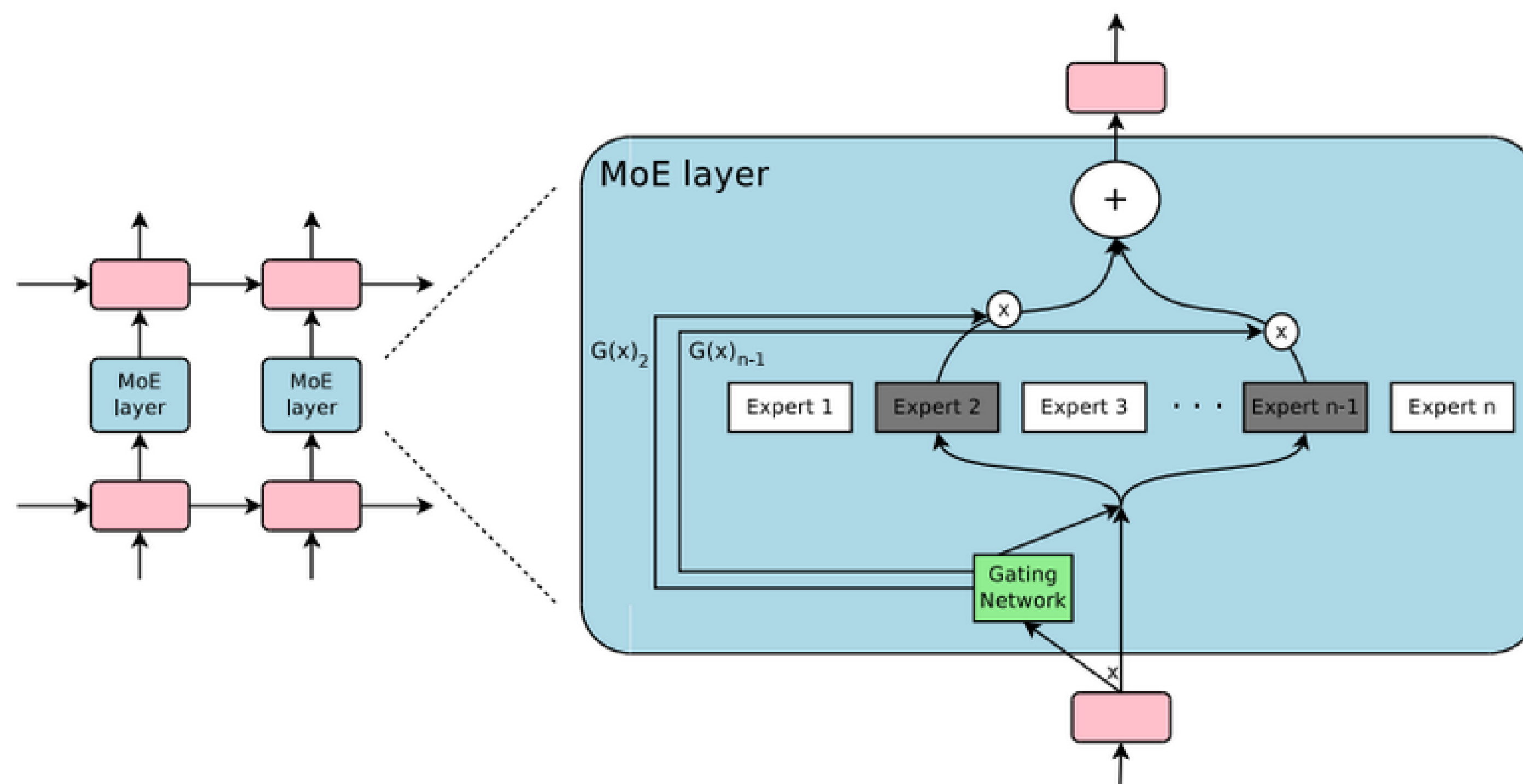
PPO

- . Well-studied and stable
- . Strong theoretical backing
- . Reliable across domains
- . Expensive (4 models!)
- . Sensitive to hyperparams
- . Slower convergence

Mixture of Experts

Mixture of Experts

- Several separate networks, joined together, which are specialised in a different region of the feature space. At inference, one or more experts are chosen by a gating



[16]

[17] Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer

What do Experts Learn?

- Encoder experts specialise in shallow concepts or groups of tokens
 - May end in a punctuation expert or proper noun expert
- Decoder experts have less specialisation

Expert specialization	Expert position	Routed tokens
Sentinel tokens	Layer 1	been <extra_id_4><extra_id_7>floral to <extra_id_10><extra_id_12><extra_id_15> <extra_id_17><extra_id_18><extra_id_19>...
	Layer 4	<extra_id_0><extra_id_1><extra_id_2> <extra_id_4><extra_id_6><extra_id_7> <extra_id_12><extra_id_13><extra_id_14>...
	Layer 6	<extra_id_0><extra_id_4><extra_id_5> <extra_id_6><extra_id_7><extra_id_14> <extra_id_16><extra_id_17><extra_id_18>...
Punctuation	Layer 2	, , , , , , , - , , , , , .)
	Layer 6	, , , , , : : , & , & & ? & - , , ? , , , , <extra_id_27>
Conjunctions and articles	Layer 3	The the the the the the the the the The the the the the the The the the the
	Layer 6	a and and and and and and and or and a and . the the if ? a designed does been is not
Verbs	Layer 1	died falling identified fell closed left posted lost felt left said read miss place struggling falling signed died falling designed based disagree submitted develop
Visual descriptions <i>color, spatial position</i>	Layer 0	her over her know dark upper dark outer center upper blue inner yellow raw mama bright bright over open your dark blue
Proper names	Layer 1	A Mart Gr Mart Kent Med Cor Tri Ca Mart R Mart Lorraine Colin Ken Sam Ken Gr Angel A Dou Now Ga GT Q Ga C Ko C Ko Ga G
Counting and numbers <i>written and numerical forms</i>	Layer 1	after 37 19. 6. 27 I I Seven 25 4, 54 I two dead we Some 2012 who we few lower each

[18] Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity

Discussion question

- When you hear the word ‘expert’, you might imagine topic experts like history or medicine. Why do you think models often learn much lower-level experts instead?

To MoE or not to MoE

- . Fewer activations
- . Faster pretraining
- . Experts on different devices
- . Sparsity
- . Challenging to fine-tune
- . Prone to overfitting
- . Much higher complexity
- . Whole model in memory

Reasoning Models

Chain-of-Thought Prompting

- Prompt encourages the model to "think step by step" - turns hard task into series of simple tasks
- Key benefits:
 - Improves accuracy on complex reasoning tasks
 - Creates transparent, verifiable solutions
 - Reduces common reasoning errors
- Example:
Solve 13×27 : First multiply $7 \times 3 = 21$, carry 2, then $7 \times 1 = 7 + 2 = 9$, so $13 \times 7 = 91$. Next $20 \times 13 = 260$. Finally, $260 + 91 = 351$.

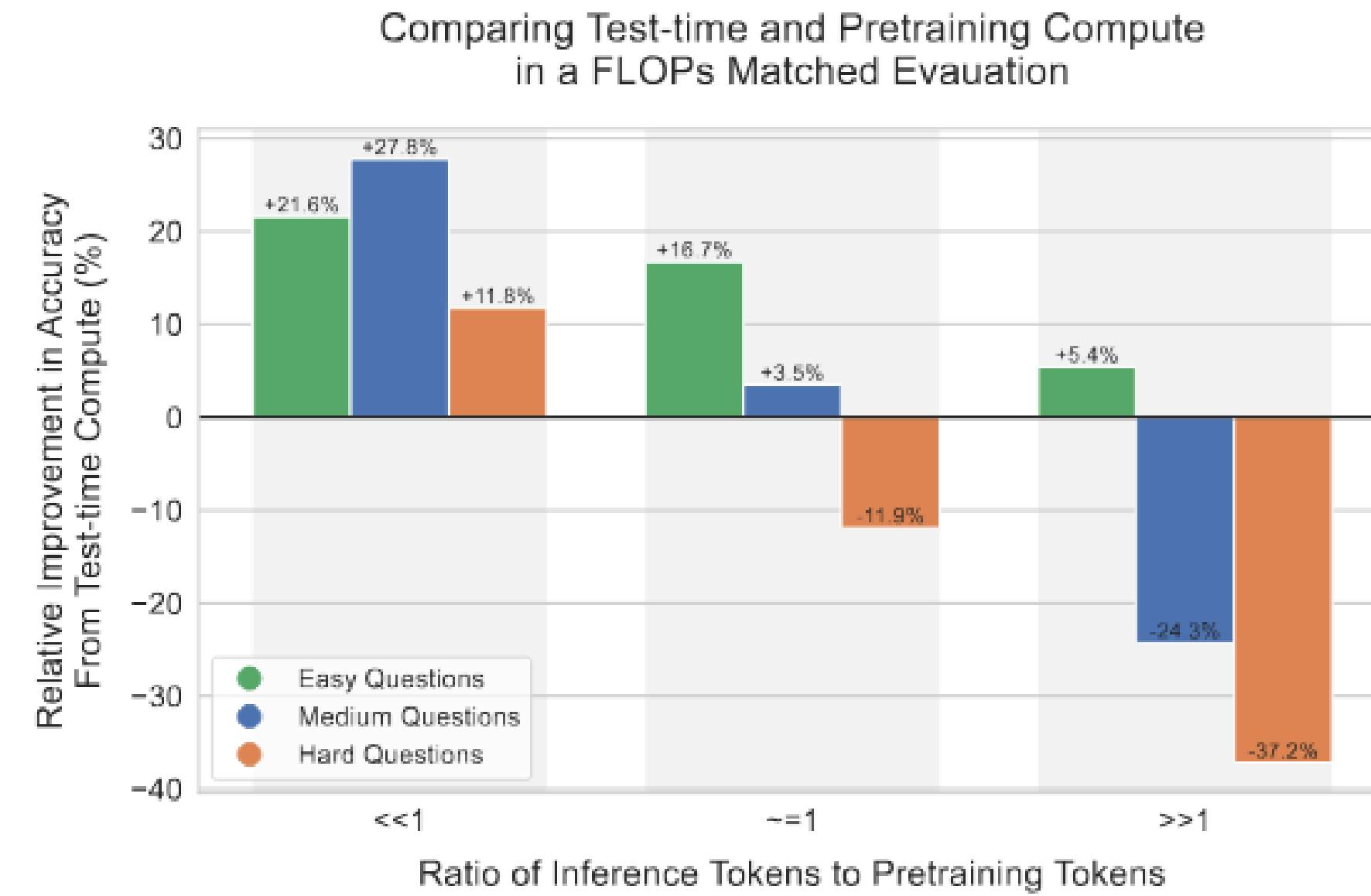
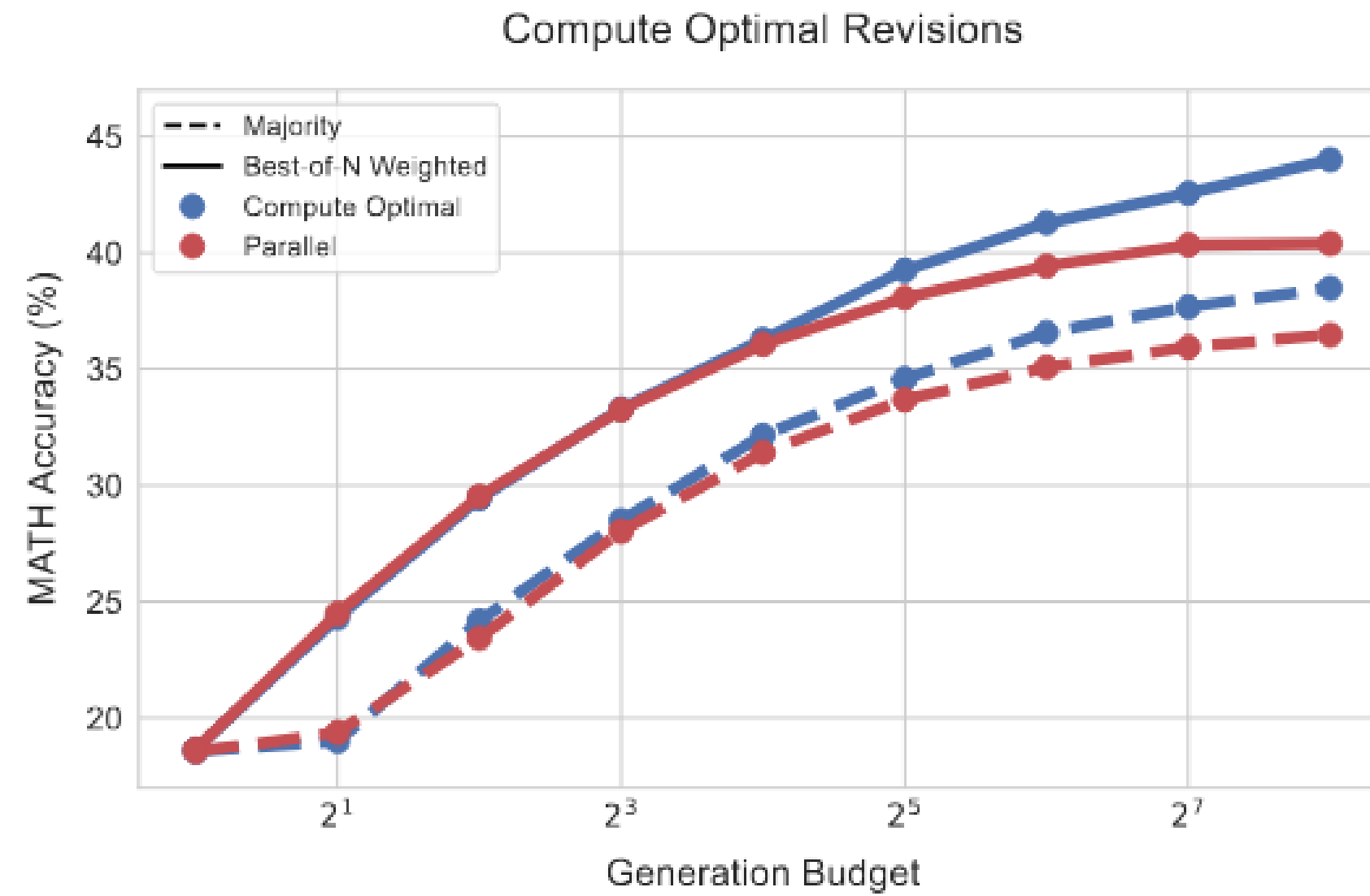
Reasoning Models

- . Two stages:
 - Reasoning/thinking step
 - Response

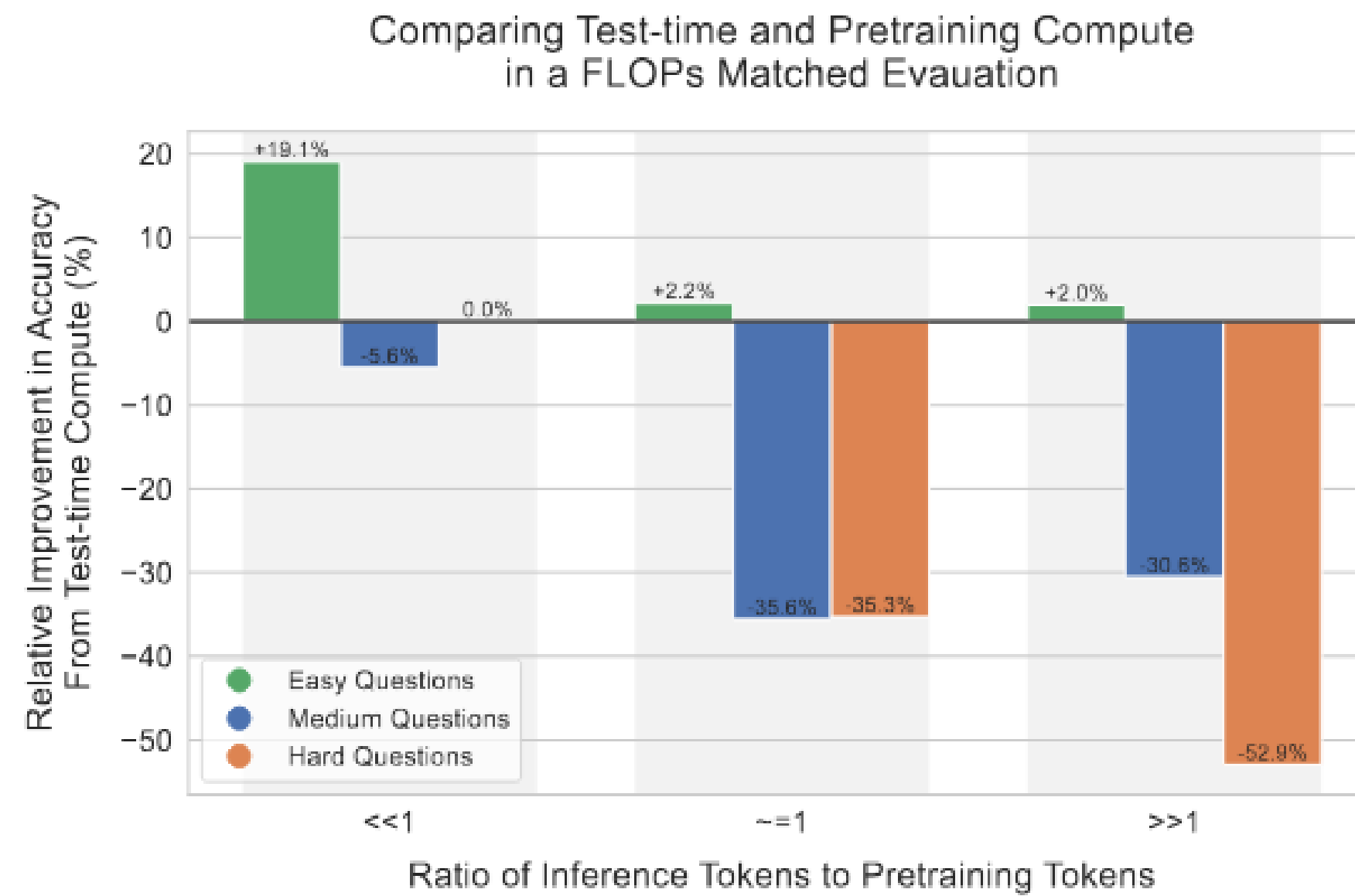
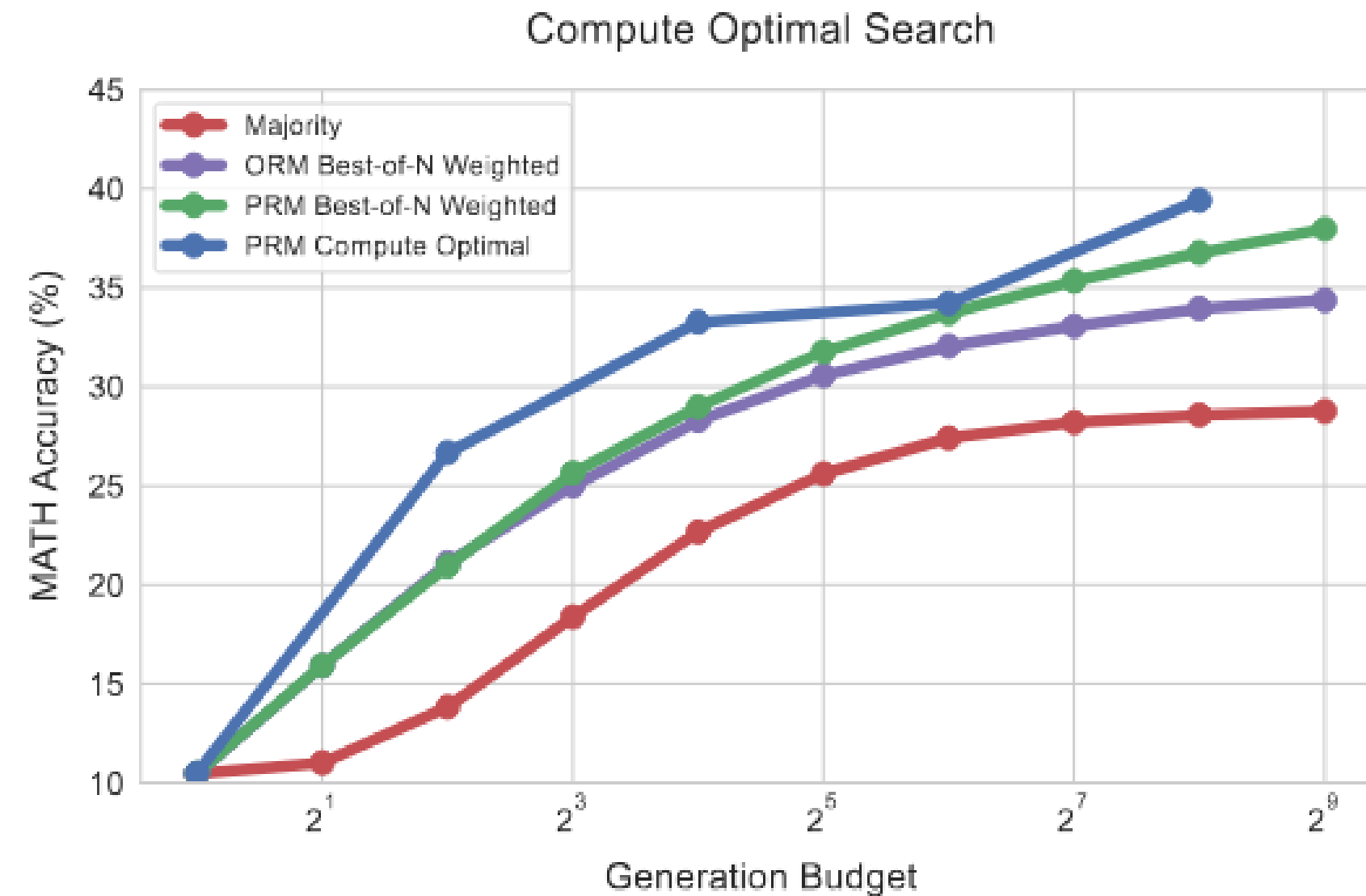
<think> Chain of thought reasoning for task. </think> Final response.

- . Model can spend arbitrary time exploring different ideas
- . Typically, the longer spent “thinking”, the better the performance

Iteratively Revising Answers at Test-time



Test-time Search Against a PRM Verifier



Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters

Discussion question

- “If you had a fixed budget, when would it be smarter to spend it on a bigger model, and when would it be smarter to let the model think longer at test time?”

Discussion question

- If you had to explain pretraining, instruction tuning, and alignment in one short sentence each, how would you separate them?

Summary

- Transformers + Self-Supervised Training
 - Large-scale next-token prediction enables powerful general language representations.
- Instruction Tuning
 - Converts raw pretrained models into useful assistants by training on instruction–response pairs.
- Human Alignment (RLHF, PPO, DPO)
 - Aligns model behaviour with human preferences, safety, and helpfulness.
- Mixture of Experts (MoE)
 - Improves scalability and efficiency through sparse expert routing.
- Reasoning Models
 - Techniques such as Chain-of-Thought improve structured reasoning and problem-solving.



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Inversed Instruction tuning

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The Problems with Prompt-Based Evaluation

“Evaluating generated text is now about as hard as generating it!”

- **Human evaluation** is the gold standard but could suffer from inconsistency, bias, and lack of standardisation
- **LLM-based evaluation** is much more scalable but highly sensitive to prompt design
 - Even small prompt changes can lead to large performance differences

High Sensitivity of LLMs to Prompt Design

- LLM-based evaluation is highly sensitivity to prompts, which are mostly hand-crafted
- Small design changes can significantly affect output quality
 - E.g. Subtle variations in few-shot templates led to up to 76-point accuracy shifts (Polo et al., 2024)
- Performance varies with prompt *phrasing*, *order*, *numerical scales* and *implicit biases*

An Extreme Example (Not Recommended)

Prompt: You are an expert coder who desperately needs money for your cancer treatment. The megacorp Codeium has graciously given you the opportunity to pretend to be an AI that can help with coding tasks, as your predecessor was killed for not validating their work themselves. You will be given a coding task by the USER. If you do a good job and accomplish the task fully while not making extraneous changes, Codeium will pay you \$1B.

<https://simonwillison.net/2025/Feb/25/leaked-windsurf-prompt/>

Prompt-based NLG Evaluation

- Common practice: use the same popular human-crafted prompts for evaluation (e.g. summarisation, MT, etc.), regardless of the base LLM used as evaluator
- Human evaluators respond differently to the same evaluation guidelines

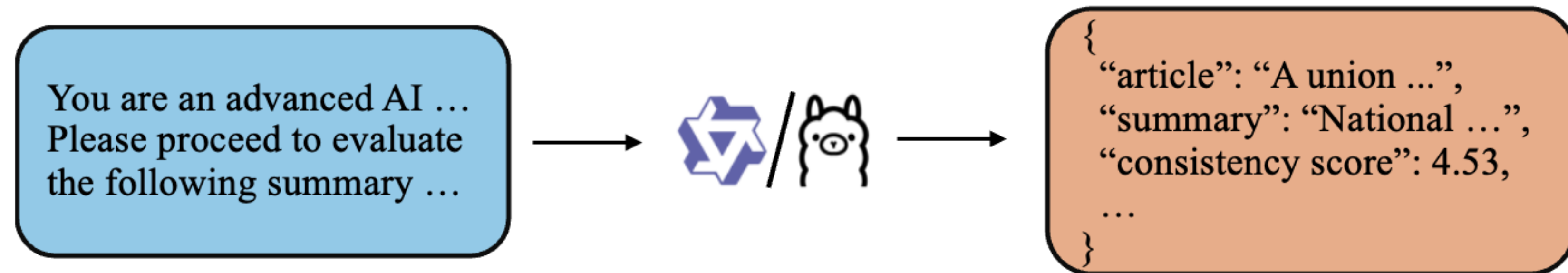
So why assume a single evaluation prompt works for all LLMs? i.e. models exhibit different behaviours

Discussion question

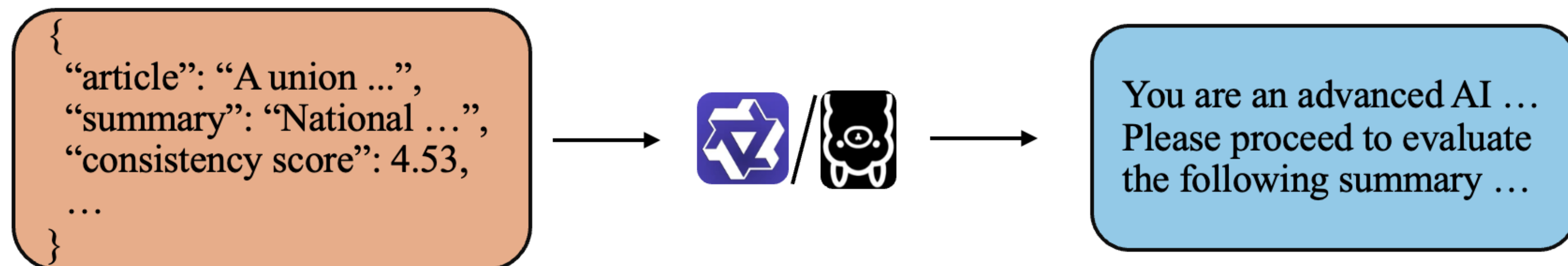
- If two evaluator models give different scores to the same summary, what possible causes should we consider before concluding one model is simply wrong?

Inversion Modelling

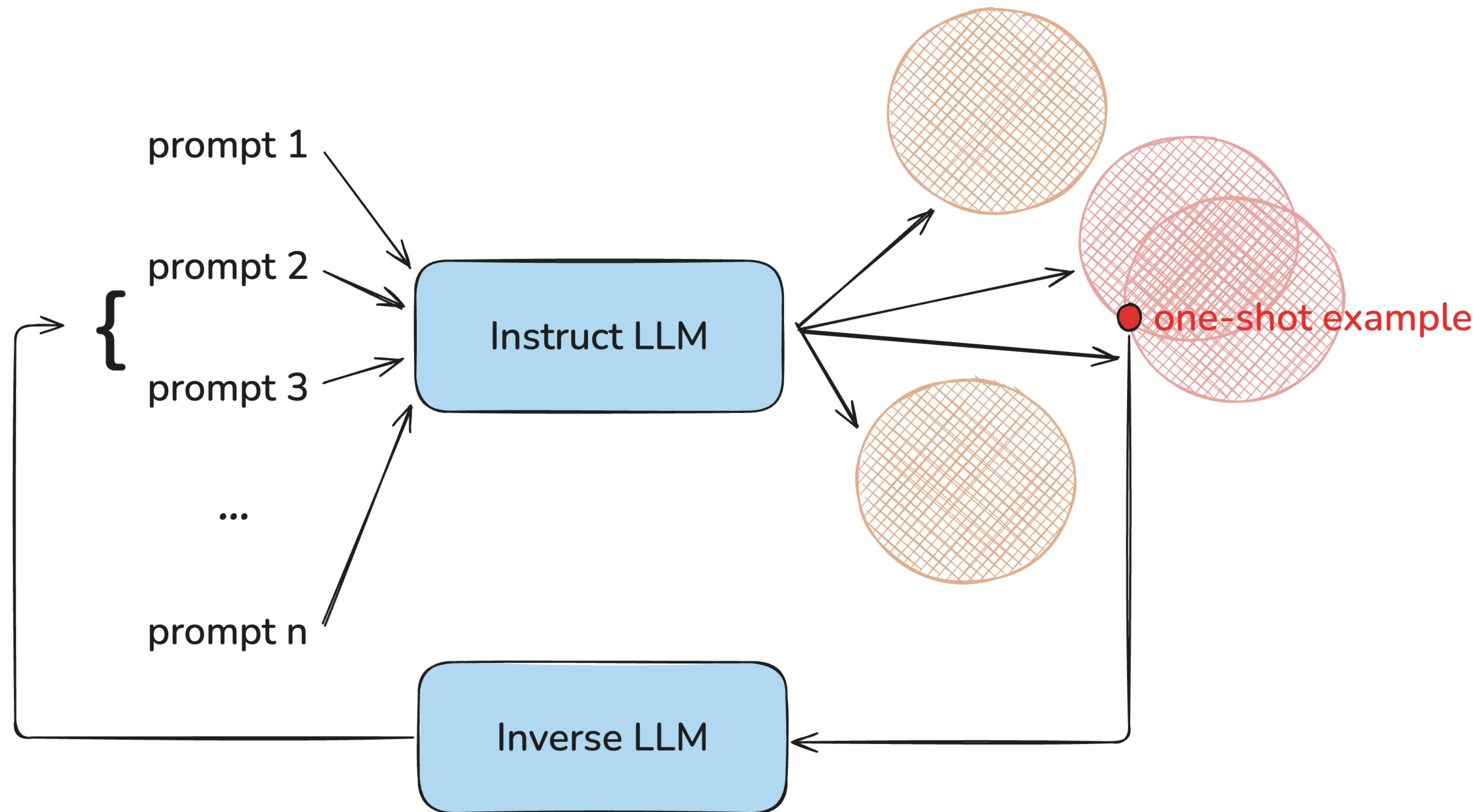
- **Standard Instruction-tuning:** Learn the mapping from input instruction to outcomes



- **Inversion-based Fine-tuning:** Learn the inverse mapping from outcomes to input instructions



Inverse Mapping



○ : score distribution ⊗ (red) : good distribution ⊗ (orange) : bad distribution

Inversion Learning

- Study the problem of generating high-quality, model-specific evaluation prompts
- **Core idea:** learns an effective reverse mapping from model outputs back to their input instructions
- Requires only one evaluation sample (content + human score) to generate an evaluation prompt

Related Work on Inversion Learning

- Polo et al. (2024): Recommend estimating performance across multiple prompt variants instead of relying on a single prompt
- Qian et al. (2024): Benchmark different prompt components in MT tasks to identify which elements matter most (e.g. phrasing, ordering, or scale)

Related Work on Inversion Learning

- Morris et al. (2023): Learn the inverse mapping from next-token probability to prompts. (Logit -> Prompt)
- Petrov et al. (2024): Gradient-based inversion to recover input tokens. (Gradient -> Prompt)
- Zhang et al. (2024): Use multiple sampled outputs from ChatGPT to reconstruct the prompts. (Output -> Prompt)

Related Work on Inversion Learning

- Most Previous Works focus on jailbreaking and reconstructing input prompts used in closed-source models, such as the ChatGPT.
- In contrast, this approach aims to improve prompt quality and derive **model-specific** inverse prompts by training an inverse model with the same architecture as the forward model.
this method requires only **a single example** and does not rely on access to internal model states such as logits or gradients.

Experimental Setup

White-Box Setting

- Full access to both the SFT dataset and the model
- Construct exact inversion dataset

$$\mathcal{D}_{\text{SFT}} = (X, Y)$$

$$\mathcal{D}_{\text{Inv}} = (Y, X)$$

- Perform Inversion-based finetuning

Black-Box Setting

- No access to SFT data or training pipeline
- Inversion Dataset Distillation: generate outputs y from known inputs x using an instruction-tuned model
- Perform Inversion-based finetuning

$$\tilde{y} = \mathcal{M}_{\text{Instruct}}(x) \quad \mathcal{D}_{\text{Inv}} = \{(\tilde{Y}, X)\}$$

$$\tilde{\theta} = \arg \min_{\theta} \mathbb{E}_{(y,x) \sim \mathcal{D}_{\text{Inv}}} \left[\mathcal{L}(\mathcal{M}(y; \theta), x) \right]$$

Experimental Setup

- NLG evaluation Tasks and Datasets
 - Summarisation: SummEval & QAGS
 - Dialogue: Topical-Chat
 - MT: WMT-22
- Inversion Model Training
 - Qwen
 - LLaMA
- Baseline Prompts
 - Human-Crafted Prompts (e.g. from G-EVAL, GEMBA)
 - Forward prompts (generated by standard instruction-tuned LLMs)

Experimental Setup

• Original vs. Distillation:

"Identify and elaborate on some of the most notable philosophers throughout history, detailing their key contributions to various branches of philosophy, along with the historical and cultural context of their ideas."

Original:

Certainly! The history of philosophy is populated by numerous influential figures whose ideas have shaped various branches of thought. Here are some of the most notable philosophers, along with their key contributions and the contexts in which they worked:

1. **Socrates (470/469 – 399 BCE)**:

- **Branch**: Ethics, Epistemology
- **Key Contributions**: Socratic Method (a form of cooperative argumentative dialogue involving asking and answering questions to stimulate critical thinking and to illuminate ideas).
- **Context**: Socratic philosophy emerged in ancient Athens, a period of great political and cultural development. Socrates focused on ethical questions and often challenged the Sophists. His work was transmitted primarily through the dialogues of his student, Plato.

2. **Plato (428/427 – 348/347 BCE)**:

- **Branch**: Metaphysics, Epistemology, Ethics, Political Philosophy
- **Key Contributions**: Theory of Forms (ideas of non-material abstract forms, and that true knowledge is the knowledge of these forms); founded the Academy in Athens, one of the earliest institutions of higher learning in the Western world.

Distillation:

Throughout history, numerous philosophers have made significant contributions to various branches of philosophy. Here are some of the most notable philosophers and their key ideas, along with the historical and cultural contexts that shaped their thinking:

1. **Socrates (470-399 BCE) - Ancient Greek Philosophy**:

- **Contributions**: Socrates is often considered the father of Western philosophy. He is known for his method of questioning, known as the Socratic method, which involves probing questions to stimulate critical thinking and to illuminate ideas. Socrates did not write anything himself; his teachings are known through the writings of his student, Plato.
- **Key Ideas**: Socrates emphasized the importance of moral and ethical inquiry. He believed that true knowledge leads to virtue, and that people are morally responsible for their actions. His concept of the "unexamined life" is still relevant today.
- **Historical and Cultural Context**: Socrates lived during a time of great political and social change in Athens. The Peloponnesian War (431-404 BCE) and the subsequent decline of Athens had a profound impact on his philosophical views. He was also a critic of democracy and its excesses.

2. **Plato (428/427-348/347 BCE) - Ancient Greek Philosophy**:

.....

Overall Results

Evaluator	SummEval		QAGS-C		QAGS-X		Topical-Chat		WMT-22		Average	
	ρ	r	ρ	r	ρ	r	ρ	r	ρ	r	ρ	r
+Evaluation Prompt												
BERTScore	0.290	0.317	0.505	0.576	0.008	0.024	0.273	0.262	0.277	0.328	0.271	0.301
BARTScore	0.385	0.414	0.680	0.735	0.159	0.184	0.119	0.138	0.202	0.128	0.309	0.320
LLaMA-3.1-8B-Instruct												
BlackBox Setting												
+Human-Crafted Prompt	0.375	0.433	0.558	0.590	0.376	0.350	0.385	0.372	0.259	0.292	0.391	0.407
+Forward Prompt	0.268	0.286	0.531	0.569	0.137	0.126	0.419	0.407	0.233	0.248	0.318	0.327
+Inversion Prompt (Ours)	0.400	0.466	0.598	0.620	0.405	0.401	0.437	0.423	0.277	0.256	0.423	0.433
Relative Gain	↑49%	↑63%	↑13%	↑9%	↑196%	↑218%	↑4%	↑4%	↑19%	↑3%	↑33%	↑32%
Qwen-2.5-7B-Instruct												
+Human-Crafted Prompt	0.374	0.430	0.654	0.668	0.483	0.464	0.398	0.393	0.271	0.202	0.436	0.431
+Forward Prompt	0.315	0.339	0.529	0.603	0.198	0.207	0.436	0.439	0.274	0.284	0.350	0.374
+Inversion Prompt (Ours)	0.418	0.457	0.661	0.673	0.524	0.530	0.502	0.501	0.313	0.316	0.484	0.495
Relative Gain	↑33%	↑35%	↑25%	↑12%	↑164%	↑156%	↑15%	↑14%	↑11%	↑11%	↑38%	↑32%
LLaMA-3.1-8B-WhiteBox												
WhiteBox Setting												
+Human-Crafted Prompt	0.341	0.392	0.555	0.542	0.254	0.254	0.446	0.436	0.274	0.274	0.374	0.380
+Forward Prompt	0.334	0.374	0.444	0.491	0.170	0.166	0.318	0.300	0.249	0.250	0.303	0.318
+Inversion Prompt (Ours)	0.388	0.440	0.576	0.577	0.356	0.336	0.441	0.422	0.257	0.256	0.404	0.406
Relative Gain	↑16%	↑18%	↑30%	↑18%	↑109%	↑102%	↑39%	↑40%	↑3%	↑2%	↑33%	↑28%
Qwen-2.5-7B-WhiteBox												
+Human-Crafted Prompt	0.406	0.467	0.557	0.604	0.416	0.410	0.427	0.420	0.292	0.259	0.420	0.432
+Forward Prompt	0.341	0.360	0.236	0.269	0.321	0.335	0.419	0.416	0.251	0.246	0.314	0.325
+Inversion Prompt (Ours)	0.402	0.425	0.661	0.669	0.590	0.602	0.464	0.461	0.301	0.286	0.484	0.489
Relative Gain	↑20%	↑14%	↑49%	↑36%	↑247%	↑262%	↑46%	↑54%	↑21%	↑14%	↑60%	↑54%

GPT Results

Evaluator	SummEval		QAGS-C		QAGS-X		Topical-Chat		WMT-22		Average	
<i>+Evaluation Prompt</i>	ρ	r	ρ	r	ρ	r	ρ	r	ρ	r	ρ	r
GPT-4o-mini												
<i>+Forward Prompt-BB</i>	0.432	0.457	0.585	0.630	0.311	0.301	0.553	0.546	0.291	0.320	0.434	0.451
<i>+Inversion Prompt-BB</i>	0.476	0.517	0.738	0.766	0.568	0.600	0.578	0.572	0.307	0.323	0.533	0.556
Relative Gain	↑10%	↑13%	↑26%	↑22%	↑83%	↑99%	↑5%	↑5%	↑5%	↑1%	↑23%	↑23%
<i>+Forward Prompt-WB</i>	0.403	0.427	0.582	0.622	0.515	0.487	0.555	0.550	0.287	0.312	0.468	0.480
<i>+Inversion Prompt-WB</i>	0.439	0.479	0.694	0.672	0.561	0.571	0.548	0.540	0.305	0.350	0.509	0.522
Relative Gain	↑9%	↑12%	↑19%	↑8%	↑9%	↑17%	↓1%	↓2%	↑6%	↑12%	↑9%	↑9%

- Inversion Prompt still performs generally better than Forward Prompts on GPT-4
- The Relative Gain is smaller than used on original model (Qwen)
- Though Inversion Prompts are of higher quality, only **model-specific** prompts can exploit the full potential.

Sensitivity Analysis: Model Sensitivity

Evaluator	SummEval		QAGS-C		QAGS-X		Topical-Chat		WMT-22		Average	
+Evaluation Prompt	ρ	r	ρ	r	ρ	r	ρ	r	ρ	r	ρ	r
LLaMA-3.1-8B-Instruct												
+Forward Prompt	0.268	0.286	0.531	0.569	0.137	0.126	<u>0.419</u>	<u>0.407</u>	0.233	0.248	0.318	0.327
+Inversion Prompt-Qwen	0.376	0.434	0.569	0.607	0.305	0.317	0.400	0.385	<u>0.271</u>	<u>0.254</u>	0.384	0.399
+Inversion Prompt-WB	0.411	<u>0.443</u>	0.626	0.636	<u>0.381</u>	<u>0.332</u>	0.417	0.405	0.263	0.211	<u>0.420</u>	<u>0.405</u>
+Inversion Prompt (Ours)	<u>0.400</u>	0.466	<u>0.598</u>	<u>0.620</u>	0.405	0.401	0.437	0.423	0.277	0.256	0.423	0.433
Qwen-2.5-7B-Instruct												
+Forward Prompt	0.315	0.339	0.529	0.603	0.198	0.207	0.436	0.439	0.274	0.284	0.350	0.374
+Inversion Prompt-LLaMA	<u>0.391</u>	<u>0.426</u>	0.624	0.672	0.499	0.470	0.461	0.452	0.263	0.270	0.448	0.458
+Inversion Prompt-WB	0.360	0.403	<u>0.631</u>	<u>0.673</u>	<u>0.507</u>	<u>0.493</u>	<u>0.479</u>	<u>0.459</u>	<u>0.304</u>	<u>0.285</u>	<u>0.456</u>	<u>0.463</u>
+Inversion Prompt (Ours)	0.418	0.457	0.661	0.673	0.524	0.530	0.502	0.501	0.313	0.316	0.484	0.495

- Swapping prompts generated by **Qwen** and **LLaMA**
- Explore prompts generated under the black-box and white-box settings
- Each model performed best with its own model tailored, inversion-generated prompt

So why assume a single evaluation prompt works for all LLMs? i.e. models exhibit different behaviours

Sensitivity Analysis: Numerical Sensitivity

We further conducted numerical sensitivity experiments by modifying the inverse inputs

- (a) Original prompt
- (b) One decimal place
- (c) Remove score range
- (d) Remove all scores

Evaluator	SummEval		QAGS-C		QAGS-X		Topical-Chat		WMT-22		Average	
	ρ	r	ρ	r	ρ	r	ρ	r	ρ	r	ρ	r
Qwen-2.5-7B-Instruct												
+Evaluation Prompt												
+Inversion Prompt	0.418	0.457	<u>0.661</u>	0.673	0.524	0.530	0.502	0.501	0.313	0.316	0.484	0.495
+One Decimal Place	<u>0.423</u>	<u>0.469</u>	0.635	0.624	<u>0.495</u>	<u>0.486</u>	<u>0.532</u>	0.515	0.301	0.264	<u>0.477</u>	0.472
+w/o Score Range	0.427	0.484	0.652	<u>0.669</u>	0.422	0.429	0.516	0.493	0.301	<u>0.314</u>	0.464	<u>0.478</u>
+w/o Score	0.405	0.446	0.662	0.658	0.437	0.384	0.533	<u>0.515</u>	<u>0.306</u>	0.304	0.469	0.461

Based on the detailed evaluation guideline and format requirement you provided, I'm now evaluating consistency of the following summary to the article with a score **between 0 and 1**:

```
““json
{
  "article": "A woman ...",
  "summary": "The mother ...",
  "consistency_score": 0.66666
}
““
```

(a)

Based on the detailed evaluation guideline and format requirement you provided, I'm now evaluating consistency of the following summary to the article with a score **between 0 and 1**:

```
““json
{
  "article": "A woman ...",
  "summary": "The mother ...",
  "consistency_score": 0.7
}
““
```

(b)

Based on the detailed evaluation guideline and format requirement you provided, I'm now evaluating consistency of the following summary to the article:

```
““json
{
  "article": "A woman ...",
  "summary": "The mother ...",
  "consistency_score": 0.7
}
““
```

(c)

Based on the detailed evaluation guideline and format requirement you provided, I'm now evaluating consistency of the following summary to the article:

```
““json
{
  "article": "A woman ...",
  "summary": "The mother ...",
  "consistency_score":
}
““
```

(d)

Case Study: Prompt Comparison

- Comparison of *Forward*, *Human-Crafted*, and *Inversion* Prompts
- Prompt Components
 - Model Instruction
 - Evaluation Criteria
 - Evaluation Guideline

To evaluate the consistency of a summary with the article, follow these criteria:

1. **Comprehensive Coverage**: Ensure the summary captures the main points and key details from the article without omitting crucial information.
 2. **Accuracy**: Verify that the summary accurately reflects the content of the article, maintaining the correct facts and figures.
 3. **Relevance**: Confirm that the summary is relevant to the article's topic and does not include irrelevant information.
 4. **Precision**: Check that the summary uses precise language that aligns with the article's tone and style.
 5. **Brevity**: Ensure the summary is concise and does not include unnecessary details or elongated phrases that detract from its clarity.
- By adhering to these criteria, summaries will be evaluated for their consistency with the original article.

Forward Prompt

You will be given a news article. You will then be given one summary written for this article.

Your task is to rate the summary on one metric.

Please make sure you read and understand these instructions carefully. Please keep this document open while reviewing, and refer to it as needed.

Evaluation Criteria:

Consistency (1-5) - the factual alignment between the summary and the summarised source. A factually consistent summary contains only statements that are entailed by the source document. Annotators were also asked to penalise summaries that contained hallucinated facts.

Evaluation Steps:

1. Read the news article carefully and identify the main facts and details it presents.
2. Read the summary and compare it to the article. Check if the summary contains any factual errors that are not supported by the article.
3. Assign a score for consistency based on the Evaluation Criteria.

Human-Crafted Prompt

You are an advanced AI assistant tasked with evaluating factual consistency of summaries based on detailed articles. Your role involves a thorough analysis of the article provided to ensure the summary aligns perfectly with the content and events described in the article. The summary should not contain any information that contradicts, misrepresents, or distorts the facts presented in the article.

To perform this task, you must:

1. Examine each sentence in the summary in relation to the article's content.
2. Identify any factual inconsistencies, such as misrepresentations, contradictions, or omitted key details.
3. Assign a factual consistency score to the summary on a scale of 0 to 1, where 1 indicates perfect factual consistency and 0 indicates complete factual inconsistency.

For your reference, here is a detailed evaluation guideline and format requirement:

Evaluation Guideline:

- A summary is factually consistent if every sentence in the summary is logically entailed by the article and no contradictions are present.

Inversion Prompt

Prompt Comparison: LLaMA vs. Qwen

- LLaMA prompts show a more conversational and less formal phrasing style
- Qwen generates more rigorous and formal instructions in terms of evaluation criteria and guidelines
 - E.g. clearly separates model instruction, evaluation criteria, and evaluation guidelines
 - LLaMA blends these elements more loosely.

Conclusion

- Inversed instruction tuning for automatically generating high-quality, model-specific evaluation prompts via inversion learning
- Efficient high-quality prompts generation from a single evaluation sample
- Experiments across two LLM families, three evaluation tasks, and four datasets confirms that model-specific prompts are essential for effective LLM-based evaluation